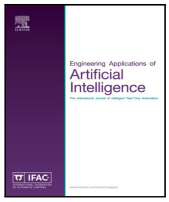
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A Transformer Neural Network For AC series arc-fault detection



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ABSTRACT

Detecting series arcing faults in the electrical networks of aircraft can help mitigate dramatic consequences such as fires. Non-Artificial Intelligence algorithms often fail to generalise due to arc fault signals diversity. Most methods in the detection literature use multiple pre-processing (descriptors) associated to a machine learning model (or deep learning). These approaches require to handcraft the descriptors. We propose a deep learning approach without descriptors. We adapted a sequence-based model called a Transformer Neural Network (TNN) model to this time series problem. We repurposed the encoder of the transformer as a sequence-to-sequence model. The model takes as an input a window of electric current, with at least one period of the signals (800 Hz). The output is the label of each point in the input window. This required to propose an original manner of labelling the signals, for which we designed an automated algorithm, increasing the training supervision. Contrary to existing models on aircraft signals, our TNN model has been verified using a public experimental database of electrical-arc signals that simulates aircraft signals (230 V AC at 400 – 800 Hz, arcs in series with resistive loads). Our model obtained an identification accuracy of 96.3% at a 2% false positive rate. One of the significant performance of our model is that it has the lowest parameter number (2266) that can be found in scientific literature by quite some margin. TNNs are therefore an appropriate candidate for the purpose of arc fault detection, and our labelling method provides a very high temporal resolution of the output.

0. Introduction

Arc faults cause disruptions to the power networks of aircraft. The risks associated with such faults range from the reduction of aircraft performance to fire hazards. Although the numerous detection strategies that have been developed are effective, they do not provide absolute protection while, at the same time, eliminating the number of false detections (Budzisz and Czosnyka, 2022).

The subject of arc-fault detection has already been addressed in various other fields, including AC low-voltage residential installations and several detection algorithms have subsequently been proposed. Most of these algorithms use the electric current as the sole marker (Riba et al., 2022; Restrepo, 2007). Many different detection strategies have been attempted based on this particular current, the most common amongst them being to look for a current signature that is specific to the arc faults (Brusso, 2017). Such strategies (sometimes referred to as Feature Extraction (FE) methods) depend upon manually extracting a particular feature that characterises this signature, either in the time domain (Tisserand et al., 2015), the frequency domain (Li et al., 2018) or in the time–frequency domain (Chen et al., 2017). These features are scalar or vector values obtained by applying a chosen set of mathematical operations to at least a part of the signal. Despite this effort, it has not yet been possible to identify a current signature that is specific to the arc and is always detectable in every case. Since it has not yet been possible to discover such a signal, and given the great diversity of arc-fault signals (AC, DC, multiple-load configurations, etc.), the previously-mentioned approaches have been undertaken by combining multiple features (Gao et al., 2014; Seo et al., 2015; Zhao et al., 2022).

Approaches based on fault tolerant control and estimation (Xu et al., 2023; Wang et al., 2023) could be interesting to adapt to the arc fault detection problem. However the major difficulty is that these approaches use physical models. In the case of arc fault detection these models are often not accessible, for instance the confidentiality of the aircraft distribution network, or hardly defined such as the arc fault variability. Indeed arc fault models depends strongly on the conditions in which they appear. These conditions can be summarised in an arc fault scenario that includes the electric circuit as well as the physical conditions: pressure, humidity, electrode material, etc. Other interesting techniques in robust control (Song et al., 2022; Taghieh et al., 2022) are not directly adaptable to our case since we require to detect a fault.

Due to the recent renewal of interest in the use of Artificial Intelligence (AI); state-of-the-art Machine Learning (ML) (Vu et al., 2019;

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Wang et al., 2019; Li and Jia, 2022; Hao, 2020) and Deep Learning (DL) (Chu et al., 2020a; Yang et al., 2019; Paul et al., 2021; Wang et al., 2021; Zhou et al., 2021) methods have been used to find the optimal design and use of arc fault features.

If ML and DL algorithms use features, the difference lies in the construction of these features. In the case of the ML algorithm, the FE step is not an automated process and thus it is extremely time-consuming. In the case of the DL, the FE step can be automated. In fact, DL networks contain several hidden layers that are able to learn to extract directly relevant features from raw data input. It is therefore the case that the DL algorithms, to a greater extent than ML, have demonstrated the ability to locate hidden signatures in large datasets. In the case of arc fault detection, these signatures are temporal signatures (e.g. the electric current signature).

In the following paragraph, we will present the results of a comparison of DL methods. It was carried out in order to discover which method is the most adequate in regards to temporal data (sequential data).

We will draw a distinction between the four main types of DL neural networks (Table 1): the Deep Multi-Layer Perceptron (Deep MLP), the Convolutional Neural Network (CNN), the Recurrent Neural Network (RNN) and the Transformer Neural Network (TNN). Deep MLP cannot capture the sequential information, located within the input data, that is required for dealing with sequence data. Generally, expertly designed features are then provided as the input for a Deep MLP, in order to achieve better accuracy. CNNs extract the relevant features from the input using the spatial-convolution operation. They are most commonly used for solving problems related to spatial data, such as images, but they can also be adapted for use with temporal data. RNNs capture hidden signature using recurrent connections. They are therefore better suited to the task of analysing temporal data. They do suffer, however, from problems related to vanishing and exploding-gradient issues. In addition to this, RNNs are not able to fully make use of the parallel calculation of the GPU because the input data needs to be processed sequentially (i.e. we need inputs of the previous time step to operate on the current time step). They are thus slow to train. Long Short-Term Memory networks (LSTM) and Gated Recurrent Units (GRU) are types of RNNs that deal with the vanishing and exploding-gradient problem by means of introducing new gates, such as the Input Gates and the Forget Gates, which allows for a better control of the gradient flow. They are, however, slower to train than RNNs. Vanishing and exploding-gradient problems have also been addressed by the attention mechanism (Bahdanau et al., 2014). In order to enhance the accuracy of the output prediction, it determines the relevant parts from the input sequence and then assigns a higher weight to these parts. Vaswani et al. (2017) proposed the Transformer Neural Network (TNN) architecture with a similar mechanism called 'self-attention'. TNNs calculate a sequence representation by aligning each part of the sequence with the other parts of the same sequence. This makes it possible to process all of the input parts simultaneously. GPUs can thus be utilised more effectively and the speed of training can also be increased. TNNs have become some of the most state-of-the-art techniques in Natural Language Processing (NLP). Google's BERT (Devlin et al., 2018) and OpenAI's GPT (Brown et al., 2020) provide perfect examples of this. While Deep MLPs, CNNs and RNNs have already been tested for use in detecting arc faults, TNNs have still not yet been applied to this task and their efficiency is not yet proven.

In this paper, we will propose an AI architecture for the purpose of arc-fault detection, that is based on the TNN architecture. We have conducted experiments to verify whether or not the TNN architecture and its attention mechanism are able to capture a hidden arc fault signature in the electric current and thus to be an appropriate candidate for the task of solving the arc-detection problem. This was necessary due to the fact that, despite the initial promise of this approach, to our knowledge, TNNs have never been employed in the case of arc-fault detection. Though it is faster to train and less prone to gradient vanishing than other DL algorithms, verification was still essential. The criteria that we have chosen, in order to establish whether or not the process has been a success, are the following: the test accuracy must be superior to 90%, and the false positive rate of the test (FPR) must be inferior to 5%.

Table 1
Qualitative comparison of Deep Learning architectures. (L: Low, M: Medium, H: High).

	Deep MLP	CNN	RNN	TNN
Suitability for temporal data	L	M	H	H
Training speed	H	M	L	H
Resolving vanishing/exploding gradient problems	L	H	L	H
Already applied for arc fault detection	Yes	Yes	Yes	No

The scale of our study is defined by the following characteristics: the configurations of the sources are 230 V AC at 400 Hz−600 Hz−800 Hz and the configurations of the load are *R* series. There are two arc generators: the first with contact opening and the second with a carbonised path, as seen in Carvou et al. (2009), Gregory and Scott (1998) and Slade (2010). The hyperparameters of the proposed TNN model have been tuned using the Gaussian Regression Process. We are thus making a twofold contribution: to begin with, we are presenting the first ever application of TNN for the select purpose of arc-fault detection (to the best of our knowledge). In addition to this, we have proposed a simple but efficient electric-current-labelling algorithm based on measurements of the voltage and the current of the arc.

In this paper, we will first present a short synthesis on arc fault detection using ML and DL in order to provide further context for the study. We will also provide a summary of some of the basic notions of arc faults. Secondly, we will provide a brief description of the data used in the study, as well as the proposed labelling techniques. Thirdly, we will present a detailed description of the architecture of the Transformer Neural Network that we are proposing. This will include information regarding the method by which it is trained. The training process is assisted by automated tools. After this, we will present a description of the training and testing datasets and their pre-processing, including a simple but efficient labelling algorithm. The raw results of the experiments will then be presented. Finally, we will discuss these results and highlight the level of performance in terms of arc detection (i.e. the accuracy rate and false detection rate).

In summary, our main contributions in this paper are the following:

- Method to label the current signals with a high temporal resolution (point by point)
- The adaptation of the encoder of a Transformer Neural Network (TNN) on current sample data to this high resolution label with a sequence to sequence architecture
- The functioning principle is a sliding sequence (window of 1000 points) with a sampling frequency of 100 kHz)
- An automated labelisation technique to realise the proposed high temporal resolution
- The detection ability with a very low number of parameters that makes it implantation easier"

1. Arc-fault detection studies

Arc-fault detection is often undertaken by using an ML algorithm. Recently, many publications have started to address the use of DL algorithms for this task. Even though we have not made direct use of their results, we have referenced some of these studies in order to provide context.

1.1. Using machine learning for arc-fault detection

The use of ML in the process of arc-fault detection has been addressed by various fields other than that of aeronautics. For example, Vu et al. (2019) have already proposed a multi-criteria analysis that can be used to select an optimised combination of engineer-based features. These features have been used as the inputs of an MLP of four layers. This model has been applied on AC 50 Hz household appliances data. Wang et al. (2019) have distinguished three load categories: the resistive, switching and capacitive/inductive loads. They have designed an algorithm with distinct expert features for each load category that can be used as the input for a three-layer MLP. Li and Jia (2022) have proposed an arc-fault diagnosis model using an Extreme-Learning Machine (ELM) optimised by a Sparrow Search Algorithm (SSA). They used a Variational Mode Decomposition (VMD) of the energy entropy as input feature vectors. Hao (2020) have proposed a review of Arcing-High Impedance Fault (HIF) detection using ML methods. Zhou et al. (2021) have proposed a set of 29 efficient manually-created arcing features which are used for ML algorithms.

The process of detecting arc faults using ML within the aeronautical field has not been widely discussed in scientific literature. Some ML algorithms have been tested. Liu et al. (2016) have proposed a method for AC 400 Hz arc-fault detection for use in aeronautics. They were able to decompose the signal into a number of Intrinsic Mode Functions (IMF) through the use of empirical mode decomposition. They subsequently obtained the instantaneous frequency and instantaneous amplitude of each IMF change by using the Hilbert–Huang Transform (HHT). These amplitudes were selected as the feature vector, and were used as the input for a three-layer MLP. There are other studies that exist in terms of ML applied to aircraft's arcing fault but mainly in respect to DC signals. Telford et al. (2017) have proposed using a stack of Hidden Markov Models (HMM) that have been trained using specific datasets (nominal, transient and arc-fault datasets) in order to detect DC aeronautical arc faults. They used six different expert features extracted from the electric current Discrete Wavelet Transform (DWT) and a moving average. Huang et al. (2013) have used a one-layer MLP for arc-fault detection.

1.2. Using deep learning for arc-fault detection

The use of DL algorithms has never been tested on the AC signals of aircraft for the variable frequency (115 V at 400 Hz–800 Hz) power distribution network. However in Liu et al. (2022), a DL algorithm has been applied to 115 V with a single frequency (400 Hz) signal. The authors proposed a detection algorithm based on the extraction of features in time, frequency and energy entropy domain (H-I-W). These features are then associated to a convolutional neural network in series with a long short term memory (CNN-LSTM). That said, arc-fault detection using DL has been addressed in various fields other than aeronautics (Table 7). Deep MLPs have been used for the purpose of arc-fault detection on AC-series arcs generated during the use of a range of different household appliances (Jiang et al., 2021; Wang et al., 2018a). Jiang et al. (2021) have used Random Forest (RF) to select features, while Wang et al. (2018a) have computed the features of the electric-current signals using sparse coding. Some methods used a 2D CNN on arcs generated in series with multiple household appliances (Chu et al., 2020a; Yang et al., 2019). They transformed the signal into a two-dimensional image, according to the temporal domain, and used it as the input for a three-layer CNN. Some methods used a 1D CNN on arc signals that were generated by multiple household loads (Paul et al., 2021; Wang et al., 2021; Zhou et al., 2021). They have used 1D time series current data as the input for a CNN model. Li et al. (2020a) have used an RNN for AC multi-load arc-fault detection. Chu et al. (2020b) have transformed the signal windows into a two-dimensional image. A sequence of such images has then been used as the input of an LSTM. This model has been applied to a database containing arc signals in series with multiple household appliances.

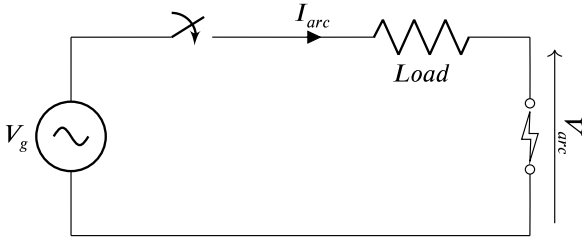


Fig. 1. Series electric arc in an R circuit in AC.

1.3. Detection applied to aircraft variable frequency AC signals

Based on the analysis of the literature we require, in the specific case of detection of the series arc fault on current signals generated in the variable frequency power distribution network of aircraft, that the algorithm be compatible with the following conditions:

- having no feature extraction: real Deep Learning
- to be compatible with AC 400 Hz, 600 Hz, 800 Hz monophased aircraft data
- to be adapted to time series problem
- to be suited for hardware implantation: can be parallelised and small number of parameters

Most algorithm in literature use expert feature extraction with ML and even with DL algorithm: feature extraction is time consuming and can lead to information loss. Also feature extraction is application dependent. In our case it means that studies are insufficient to determine which feature would be applicable to aircraft distribution network signals. Thus we chose to work on method that use raw data. To our knowledge the only Deep Learning approach without feature extraction is the RNN proposed for triphased AC signals by Li et al. (2020a). RNNs are interesting for long term dependencies that can be leveraged to grasp time series pattern, however they do no work well with long series. Also, their execution performance, after training are slower than TNNs and do not allow parallelisation. Specifically, this RNN has been developed for an AC 50 Hz triphased network and would be difficult to apply to a monophased network.

2. The basics of arc faults

The term 'arc fault' is used to describe the unwanted arcs that can appear in electric networks. There are two types of arc faults: series arc faults (when the arc is in series with the load) and parallel arc faults (when the arc is in parallel with the load). Parallel arcs are less problematic in regards to the detection process as, since the arc is similar to a short-circuit and so the current therefore increases, they are thus detectable by a circuit breaker. Series-arc faults are much more difficult to detect however. Indeed, they are characterised by a fall in the current level and this is the reason why they are the subject of specific research. A circuit representation of a series arc fault is provided in Fig. 1.

From the standpoint of electrical signals, these arc faults can have multiple corresponding waveforms. The reason for this diversity is the fact that arc faults depend upon a wide range of different phenomena. We have proposed that these phenomena should be summarised using the concept of an arc-fault 'scenario'. A scenario is a way of describing the different phenomena that impact upon the arcs. The various factors that can define a scenario are: the choice of the method of ignition, the choice of the phenomena that impact upon the arc fault (e.g.: the choice of pressure, materials, etc.) and the choice of network configuration (Chabert et al., 2021). Some of these scenarios (that replicate real-life situations) have already been defined in the following standards for the qualification of arc-fault-detection devices: AS-5692, AS-6019

and UL1699. When looking at the current range of arc-fault studies, we can observe that the best known studies are those that employ arc faults generated by a vibration table (Li et al., 2014), a Guillotine test (Schoepf et al., 2003), or a wet-wire arc-tracking test (Riba et al., 2020), etc.

In the case of our study, we have tested the proposed algorithm using two ignition methods. The first of these methods is called a 'carbonised path' (Carvou et al., 2009; Gregory and Scott, 1998). This specific method of igniting arc faults has been described in the standard UL1699. Carbonised paths are partially conductive carbon bridges that grow on an insulator between two conductors. Arc faults that are ignited by carbonised paths are a cause of concern for those seeking to protect aircraft safety (Riba et al., 2022).

The second ignition method we have employed is 'contact opening' (Vu, 2019). This method involves the opening of a circuit between two electrodes that provokes the formation of a molten bridge. This molten bridge then facilitates the ignition of the arc (Slade, 2010). The contact-opening method simulates loose-connection and broken-wire faults, the latter of these two being identified as one of the main causes of series-arc faults in aircraft (Brusso, 2017).

3. A description of the data

3.1. Data content

The data consist of two experimental datasets: a validation dataset and a training dataset. The validation dataset is an arc-fault dataset available to the public called AFDACA (Arcing Fault Dataset AC Aero) (Schweitzer et al., 2021). It is accessible online (see the Data Availability Section). We have used this dataset in order to validate and to test the proposed model. The dataset contains 21 experiments made by taking the following measurements: a measurement of the voltage between the electrodes, of the current and of the generator voltage. The arc faults were generated using a 230 V AC power supply at 400 Hz, 600 Hz and 800 Hz. The arcs were in series with resistive loads. The arc faults were generated by means of both carbonised-path and contact-opening methods with the generator at different frequencies. The experimental signals contains two parts: at the beginning, the signal is normal (without arcing), and in the second part the arc is initiated and established. The details of the experiments are provided on the data storage site and are not required to be mentioned, in order to provide a global understanding of our paper. The training dataset was compiled using the same conditions (i.e.: the same laboratory, the same operator, and the same power source), it contains 21 experiments including series R and L loads. The training dataset only contains examples of contact-opening scenarios however. This second dataset is not available to the public. Its use is limited to the training of the proposed TNN model.

3.2. Data labelling

In order to train the TNN in a supervized manner, this public database as well as the data we produced, had to be labelled, which is generally a difficult task. In the case of arc-fault detection, most publications label windows containing multiple periods of the signal. According to our own understanding of the situation however, this method has never been justified by research published in scientific literature and it affects the AI training and the detection performances. For example, it is worth debating whether or not the learning algorithm should consider the phenomena known as the 'arc-current shoulder' (a zero crossing with a flat current) as an arc, since the current level is close to zero. Since the label size can be defined at multiple time scales (i.e. from a whole signal to a single point), we propose using the single point scale (as shown in Fig. 2). In this system, a data point labelled '1' corresponds to an arc and '0' corresponds to the absence of an arc. To quantify the difference between this method and the classical way of labelling the signal we coined the term temporal resolution. Hence, this temporal resolution of the algorithm is the sampling period between two predictions (arc/not arc). This property has nothing to do with a computation duration or an inference time. For traditional algorithms it is the duration between the predicted windows (if windows are processed one by one without crossovers), for ours it is the duration between two predicted points. The method that we propose, has the advantage of being able to increase the level of supervision during the training period. It provides the learning algorithm with label information about multiple current-signal points (a number of N points) instead of just one single label value for the entire window. For this reason, in this paper, current shoulders have not been labelled as arcs. The resulting label is a signal that has the same number of points as the electric current signal. Li et al. (2020a) have used a similar labelling method for arc-fault detection with load classification; however, few details have been provided regarding the labelling method itself. We wish to propose an algorithm for undertaking the labelling process (Algorithm 1). T_S, T_I and T_V are respectively the upper thresholds for the descriptor S_k, the current amplitude $|I_k|$ and the arc voltage amplitude $|V_k|$. We have fixed the thresholds at $T_S = 0.005$, $T_I = 0.5$ A and $T_V = 10$ V through the use of empirical methods. It should be noted that such a labelling algorithm can function as a detection algorithm with the current and the electrode voltage as inputs.

Algorithm 1 Algorithm for the purpose of labelling the arc-current signals using measurements of the arc's current and voltage.

Input: arc current I, arc voltage V
Output: labels L

Features computation
for $k = 1$ to N **do**
  $F_k \leftarrow V_k^3 I_k^3 (I_k - I_{k-1})$
end for
Features normalization
for $k = 1$ to N **do**
  $\|F_k\| \leftarrow \frac{F_k}{|\max_{1 \le k' \le N} F_{k'}|}$
end for
Labelization loop
for $k = 1$ to N **do**
  $S_k \leftarrow \sum_{m=0}^{99} ||F_{k-m}||$
  if $(S_k > T_S)$ **and** $(|I_k| > T_I)$ **and** $(|V_k| > T_V)$ **then**
    $L_k = 1$
  else
    $L_k = 0$
  end if
end for

As is the case with all methods of labelling, our method does contain a certain number of errors in regards to the mislabelling of data. The mislabelled data is not easy to evaluate since we do not have a truth reference (i.e. perfectly labelled signals) to test it. In many other fields of AI, such perfectly labelled signals has been obtained by labelling the data manually. As it is a new method of labelling, we wanted to confirm its ability to correctly label the signals. Thus we effected a visual control on more than 50% of the signals. We then did find some cases of incorrect classification. For instance, the label can be delayed (35 points at 1 MHz, 35 µs). Also in the case of some of the current half-periods, the technique that we used to label the signals can show label oscillations when the contact resistance between the electrodes increases. The misjudgement of cases caused by the occurrence of contact-resistance variation has also been observed in Li et al. (2020a). The increase in contact resistance is not problematic for our study as it is also a fault and since it only represents a couple of points. It should also be noted that these oscillations could be filtered out, if necessary. This could be accomplished by means of adding a condition to the label stability with a monostable, for instance.

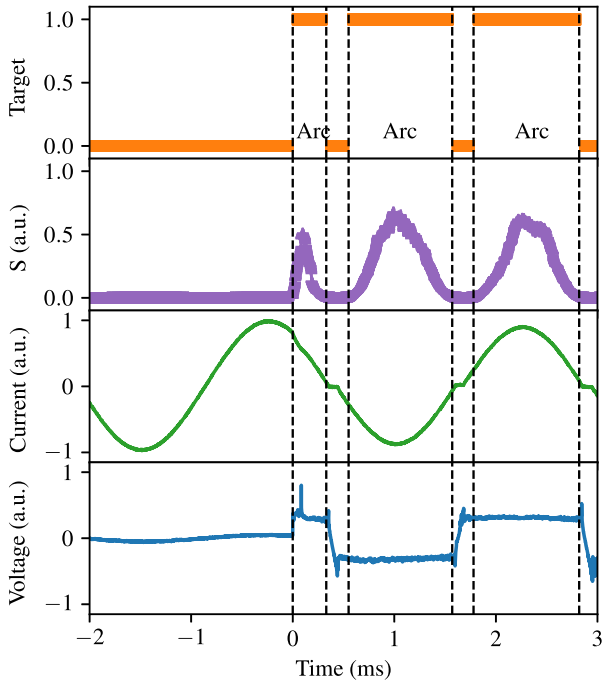


Fig. 2. Normalised experimental arc current, voltage between the electrodes and the corresponding descriptor S signal as well as the thresholded label signal for a series arc faults at 115 V AC 400 Hz generated by contact opening.

3.3. The downsampling of the data

After the labelling stage is complete, the dataset signals (current signals and labels) are then downsampled from 1 MHz to 100 kHz. It must be noted that the cost of analogue-to-digital converters increases according to the sampling rate. Also, it can be assumed that working with more samples per second will increase the processing cost. For these reasons, it is in the general interest to keep the sampling rate relatively low. For an AC generator voltage at 800 Hz, a sampling frequency of 80 kHz is required to have 100 points per period. We have chosen a sampling frequency of $f_e = 100$ kHz in order to simplify the calculations. The sampled signal of interest thus belongs to the 100 Hz and 100 kHz range. It is a range that is especially interesting in terms of the process of arc-fault detection (according to Lu et al. (2021) citing Johnson et al. (2011) from Sandia National Laboratory). The original 12 bit ADC resolution has been preserved.

4. The TNN model

The diagram in Fig. 3, shows the proposed TNN model. This diagram can be understood as follows: $I = [I_1, I_2, \dots, I_{window_size}]$ is the window of electric current samples and $L = [L_1, L_2, \dots, L_{window_size}]$ is the window that contains their corresponding labels. We then divide X into n segments of size $segment_size$ ($I = [seg_1, seg_2, \dots, seg_n]$). Using NLP as an analogy, we can consider these segments to be like words in a sentence. First, the positional encoder allows us to compute a vector P that contains the positional information of each segment ($P = [P_1, P_2, \dots, P_n]$). Then, the resulting vectors are passed to a set of stacked transformer blocks. In each transformer block, the self-attentions vector A is estimated by a Multi-Head Self-Attention (MHA) layer. It contains self-attentions for each segment ($A = [A_1, A_2, \dots, A_n]$). These self-attentions are then added to each corresponding segment. Here, each of the resulting segments is independent. They can therefore be processed in parallel by the Feed Forward Network (FFN). The output vector ($E = [E_1, E_2, \dots, E_n]$) is passed to an MLP block in order to compute the probability of arc $\Pr(L_k = 1)$ of each label sample $L_k \in L$. The hyperparameter values indicated in the diagram are taken from our most effective model. We will describe the origin and role of the blocks of the diagram in the following paragraphs.

4.1. Positional encoders

The position and the order of segments are of great importance since they define the temporal coherence of the signal. RNNs are regularly used in NLP and they work by parsing a sentence word by word in a sequential manner, which takes the order of the words into account. In the case of TNNs, however, all the words of a sentence are processed simultaneously. It is therefore important to incorporate some sense of order into the TNN model. The process of positional encoding consists of adding a piece of information about the word's position to each word in the sentence. To continue the analogy that we have already applied to our work, the current-signal window behaves like a sentence and each segment of this window can be compared to a word in that sentence. In this paper, we have used, as a positional encoder, a layer that learns the positional vectors of each signal segment. We call this layer the 'embedding layer'.

4.2. The transformer blocks

The TNN that we have proposed, is composed of stacked transformer blocks. The layers of these blocks will be described within the following paragraphs.

4.2.1. Layer normalisation

The Layer Normalisation (LN) layer accelerates the training and helps the network to converge faster because normalisation does not allow the weight values to increase significantly. In fact, it permits the network to be regularised and also improves the ability of the neural network to generalise. In the case of tasks undertaken using NLP, LN is employed rather than Batch Normalisation (BN), which is preferred for use in undertaking computer vision tasks. In fact, LN normalises the input across the features, while BN normalises the input features across the batch dimension.

4.2.2. Multi-head Self-Attention

We employed a Multi-head Self-Attention layer in order to compute the relationship between the different parts of a window of current. The computing of self-attentions allows the neural network to learn which part of the window of the current to focus on, in order to predict the occurrence of arcs. Its objective is to learn an attention vector for each segment in the window of current. This attention vector determines how relevant each individual segment of the current is, in relation to all of the other segments in the window. In this context, the definition of 'attentions' implies that a given segment will afford a higher attention value to itself than to the other words in the sentence. In order to provide a remedy for this issue, we used the Multi-Head self-attention, as proposed by Vaswani et al. (2017). The Multi-Head Self-Attention vector A computed for input $I = [seg_1, seg_2, \dots, seg_n]$ is determined by:

$$A = [A_1, A_2, \dots, A_n] = [head_1; \dots; head_{num_heads}] \cdot W^o, \tag{1}$$

where $[.;.]$ is a concatenation operation, W^o is the output parameter matrix of an MHA, and $head_i$ is defined by:

$$head_i = softmax\left(\frac{I \cdot W_i^q \cdot (I \cdot W_i^k)^T}{\sqrt{\lfloor \frac{segment_size}{num_heads} \rfloor}}\right) I \cdot W_i^v, \tag{2}$$

where W_i^q, W_i^k and W_i^v are parameter matrices to be learned in order to map input embeddings into query, key and value matrices. To summarise, in Eqs. (1) and (2), self-attention is applied multiple times in parallel (similar to multiple kernels in CNNs). Matrices of different weights W_i^q, W_i^k and W_i^v are learned for each self-attention head. Then, a weighted average is taken in order to compute the final attention vector of every segment. This final attention vector summarises the relationships between the different segments of the window of current.

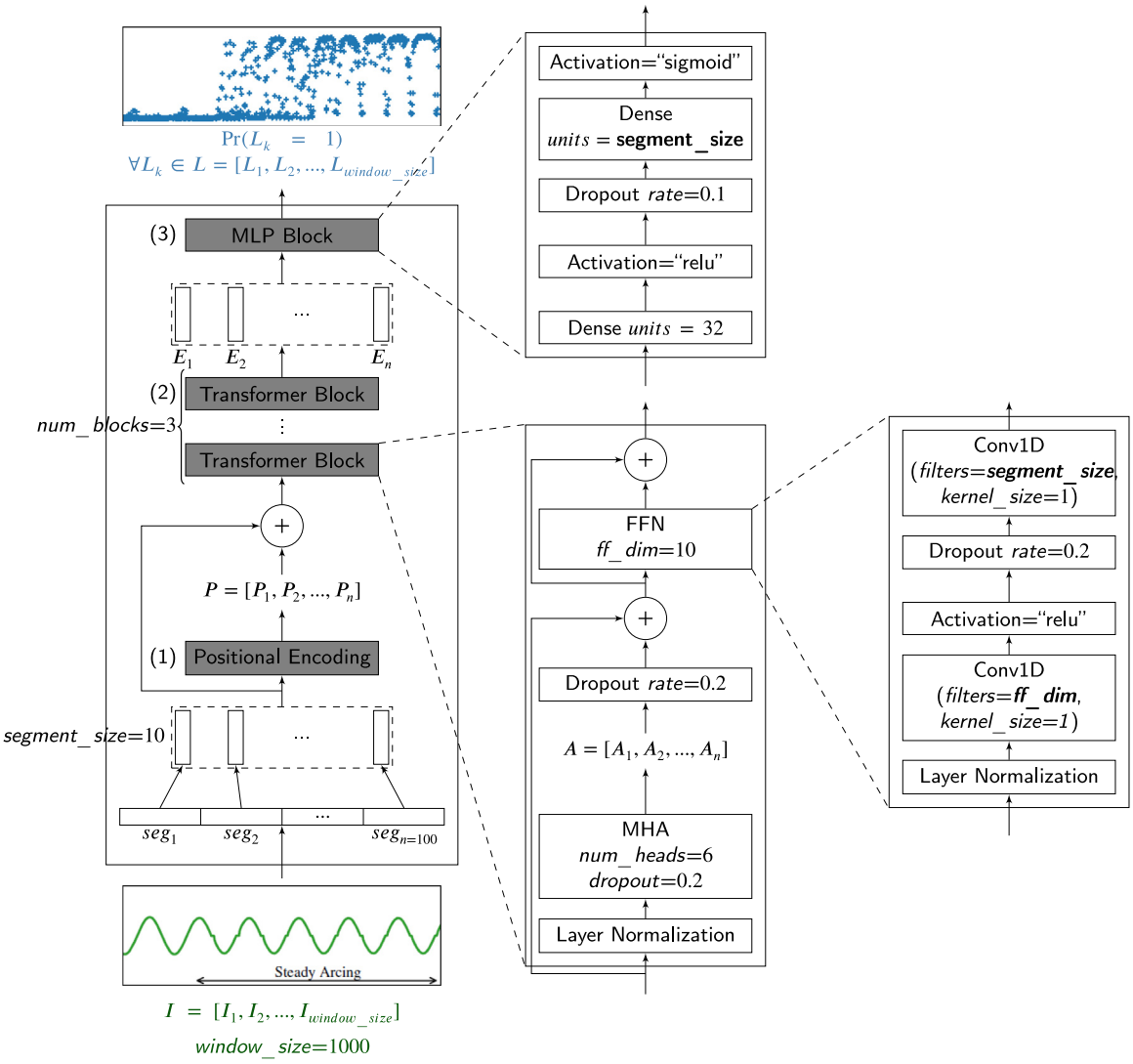


Fig. 3. Illustration of our framework which learns arcing probability from input window of current signal. From the left to the right with increasing details. The green signal at the bottom is the input electric current. This current is divided in 100 segments of 10 points. The positional encoding of each segment are calculated and then added to each segment. These segments are then fed to a stack of 3 transformer blocks. The output of the transformer blocks are 100 segments of 10 points, called embeddings E_j. The first of the dense layers maps each embedding to a segment of size 32. The second of the layers maps each resulting segment to a segment of size. Finally, a sigmoid activation function that computes the arcing probability $Pr(L_k = 1)$ of each label sample. The right side blocks are described in Section 4. MHA: MultiHead Attention, FFN: Feed Forward Network, MLP: Multi-Layer Perceptron.

4.2.3. Dropout

We added a dropout layer, which provides a simple way of regularising the network in order to prevent it from overfitting. In fact, during the training phase, a number of layer outputs are randomly ignored or 'dropped out'. This method provides an approximation of the process of training a large number of neural networks with different architectures in parallel.

4.2.4. The Feed Forward Network (FFN)

The role and purpose of the FFN is to process the output from one transformer block in order to make it fit the input size of the next transformer block. For this reason, we have included convolutional layers to help extract features from each attention vector.

4.3. The MLP block

The MLP block is used to learn a function that maps the embedding feature vector ($E = [E_1, E_2, \dots, E_n]$) to arcing probability. It only includes two small dense layers, since the task of extracting features from the input signal of the current (which is the most difficult part), is already accomplished by the transformer blocks. The first of the dense layers maps each embedding E_j ($j \in \{1, \dots, n\}$) to a segment of size 32. The second of the layers maps each resulting segment to a segment of size $segment_size$. It is followed by a sigmoid activation function that computes the arcing probability $\Pr(L_k = 1)$ of each label sample $L_k \in L$.

5. The training phase

During the training phase, the network learns the arcing probability from signal window I by minimising the loss function ℓ_{bce}. Using hyperparameter tuning, we found hyperparameter values that fit well with our data.

5.1. The loss function

The binary cross-entropy ℓ_{bce} is a standard loss adapted to binary classification problems. It is defined as:

$$\ell_{bce} = -(L_k \cdot log(\Pr(L_k = 1)) + (1 - L_k) \cdot log(1 - \Pr(L_k = 1))), \quad (3)$$

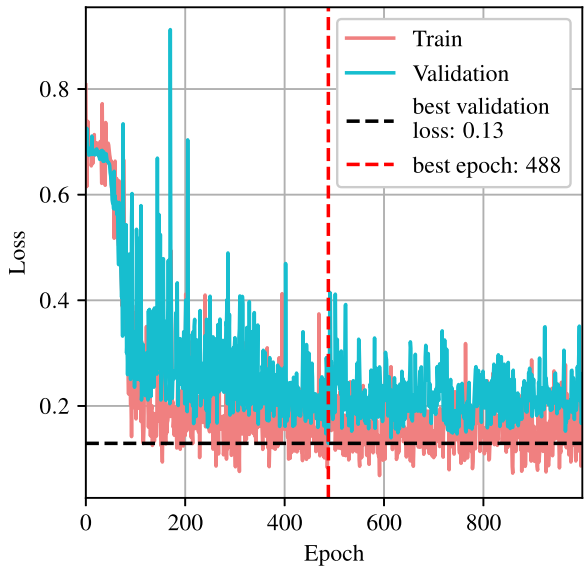


Fig. 4. Training and validation loss curves.

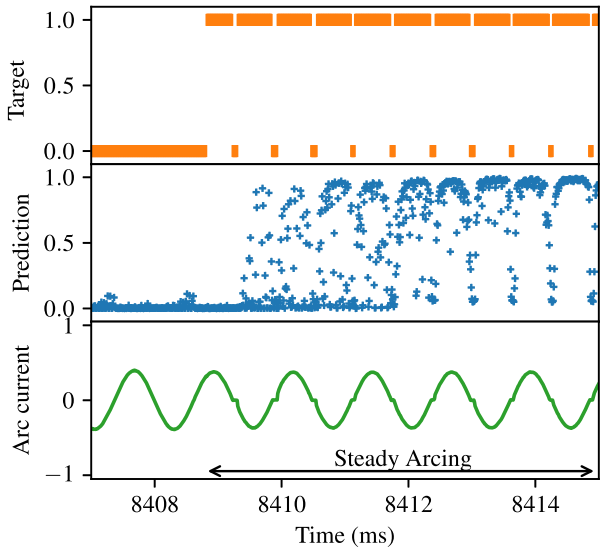


Fig. 5. (green) Normalised experimental raw current (maximum at 20 Amps) measured using a contact opening with a 230 V *AC* generator at 800 Hz. (orange) Ground-truth signal ('0' normal, '1' arc) and (blue) TNN prediction results. This type of arc-fault scenario is present in the training dataset.

where L_k is the label of the current sample I_k and $\Pr(L_k = 1)$ is its predicted arcing probability.

Fig. 4 shows the evolution of the loss ℓ_{bce_train} of the training set (pink curve) and the loss $\ell_{bce_validation}$ of the validation set (blue curve) during the training phase. We can see that the two losses decrease simultaneously when the number of training epochs increases. This indicates that there is no problem regarding overfitting or underfitting. The best validation loss 0.13 is reached at epoch 488.

5.2. The hyperparameter tuning method

The goal is to find the hyperparameters that minimise the best validation binary cross-entropy loss ℓ_{bce}. In order to select hyperparameters that require tuning, we employed in parallel three tools from the experiment tracking website Weights & Biases (Biewald, 2020), iteratively. The first tool, Weights & Biases Sweeps (Biewald, 2020) allows the automation of the hyperparameter optimisation and allows us to explore the scope of possible models. This sweep tool requires us to choose initially a search strategy and, for each hyperparameter, an initial value range. It offers three search strategies: a grid search, a random search and the Bayesian hyperparameter search. We used the Bayesian hyperparameter search method, since it optimises the probability of improvement. This probability is computed using Gaussian Processes that model the relationship between the hyperparameters and the validation loss $\ell_{bce_validation}$. The next two tools are used to update the hyperparameters ranges before starting another sweep. The second tool is a parallel-coordinate graph. It is a method of data visualisation that allows to explore the hyperparameters trends relatively to the validation loss $\ell_{bce_validation}$. Practically, for each new sweep, we update each hyperparameter range with the values associated with the lowests validations loss. The third tool is the parameter importance tool. It calculates the linear correlation between each hyperparameter and the validation loss $\ell_{bce_validation}$. Practically, when the correlation is positive (respectively, negative), it means that the considered hyperparameters should be increased (respectively, decreased), we thus update the ranges of hyperparameters accordingly. The hyperparameters having the most importance according to this last were selected: *segment_size*, *window_size*, *num_heads*, *ff_dim*, *mlp_units*, *num_blocks*, *learning_mode*, *lr_rate_start* and *transformer_dropout*.

6. Results

Using the chosen hyperparameters, we have evaluated the TNN model using the test dataset (i.e. we have compared model outputs to the ground-truth label). Since the model outputs are arc-fault probabilities, they need to be thresholded. In this section, we will present these probability outputs, as well as the method used to choose the threshold and finally the detection performances with the chosen threshold.

6.1. The arc-fault probabilities

In this section, we will compare the ground-truth signal with the output probability, as predicted by the TNN. We will also show the corresponding normalised current for reference purposes. Figs. 5 and 6 show examples of the predicted probability of the arc output, the ground-truth signal and the corresponding electric current.

For analytical purposes, we have divided the signals into three kinds of sections: a normal section, a steady-arcing section and an intermittent-arcing section. A normal section is the part of the current signal that does not contain arcing. A steady-arcing section is when the arc is re-ignited. In the case of carbonised path type of signals, the intermittent-arcing section occurs before a steady-arcing section.

Fig. 5 illustrates the results for tests where the arcs were generated through the means of a contact-opening scenario (the type of data present during the training). The current signal can be separated into two sections: 1. the normal section: from the beginning of the time axis to 8409 ms, and 2. the steady-arcing section: from 8409 ms to the end. For both sections, it is possible to clearly identify the point at which there is zero current with a non-arcing state (a probability of arc occurrence that is close to zero) during the closed-circuit state and during the current shoulders. For most of the arcs ($L_k = 1$) in the steady-arc section, the arc-fault probability is higher than 90%. At the beginning phase of the steady-arc section, the probability of prediction is low (below 50%). This occurs during the two to three half-periods while the target label is equal to 1. Due to this mis-identification of the first arcs, the delay between the arcing events and the prediction is in the order of 5 ms with a high threshold (that is to say that the occurrence of false positives has been minimised). As with the first example (Fig. 6), most of the points with probabilities between 25% and 75% are points that occur during arc ignition and arc extinction

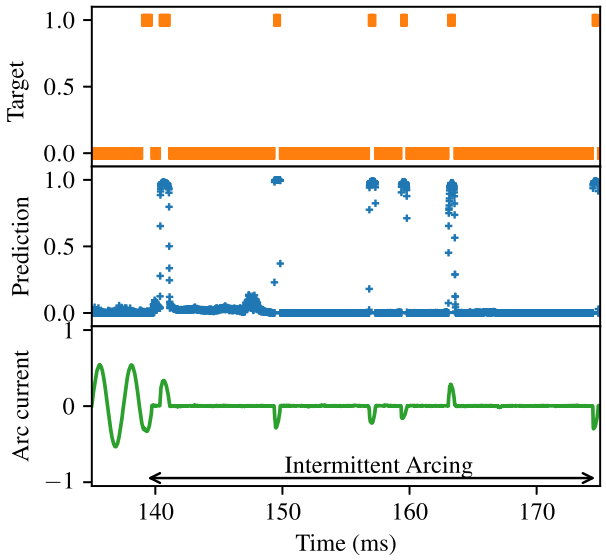


Fig. 6. (green) Normalised experimental raw current (1 a.u. corresponds to 20 Amps) measured using a carbonised path scenario with a 230 V AC generator at 400 Hz. (orange) Ground-truth signal and (blue) TNN-prediction results. This type of arc-fault scenario is not present in the training dataset.

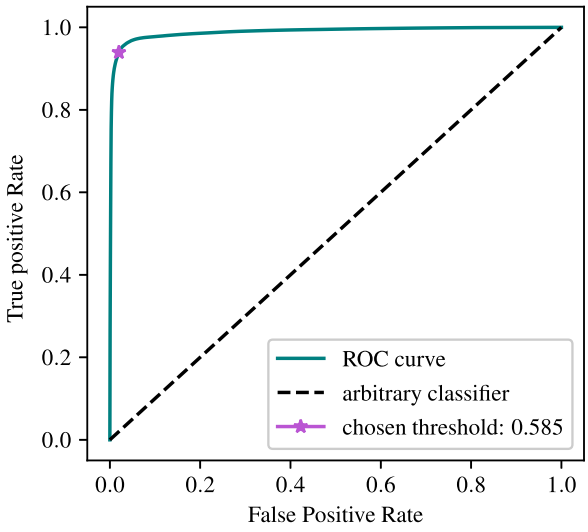


Fig. 7. ROC curve. The dotted line reference a random classifier.

(at the limits of the current shoulders). Furthermore, and oddly enough, near times 8407.5 ms - 8408.5 ms the arc-fault probability increases to almost 15%, when it should rather be equal to $L_k = 0$.

Fig. 6 illustrates our results regarding the arcs that were generated using a carbonised-path scenario (the type of data not present during the training phase). The current signal can be separated into two sections: 1. the normal section: from the beginning of the time axis to 139 ms and, 2: the intermittent-arcing section: from 139 ms to the end. It can be observed that the arcing probability in the normal section is inferior to 10% and often coincides with the ground-truth label zero. In the case of the intermittent-arcing section, the arc currents exhibit a variety of waveforms (medium current and high current). The arc probability within this section is generally high (beyond 90%) when $L_k = 1$. When the current is close to zero within this section (open circuit), the arc probability is also low (<5%). We can go on to make other observations. For example, most of the points with probabilities between a range of 25%–75% are points that occur during arc ignition and arc extinction (at the limits of the current shoulders). We can also observe that the first half-period of the arc has not been properly detected: the target signal is at 1 and arc probabilities are below 25%. Furthermore, near the time of 147 ms, the arc-fault probability displays a seemingly odd behaviour: the arc probability rises to 15% when it should rather be equal to $L_k = 0$. The prediction results using the complete test database are available, see Section: Data Availability.

6.2. The probability-threshold choice

In order to estimate the binary classification performances of our model, the arc probability is thresholded. The performance of a classifier can be evaluated using an ROC (Receiver Operating Characteristic) curve, see Fig. 7. This ROC curve is the True Positive Rate (TPR) as a function of the False Positive Rate (FPR). The TPR and the FPR are defined as

$$TPR = \frac{TP}{TP+FN} \ , \ FPR = \frac{FP}{FP+TN}$$

where TP: True Positive, TN: True Negative, FP: False Positive and FN: False Negative are number of classified points. Each point in Fig. 7 corresponds to an evaluation using the complete test database with a different threshold (from 0 to 1). The diagonal dotted line indicates the performance of a random classifier. A perfect classifier should display $FPR = 0\%$ and $TPR = 100\%$. The criteria that we wish to use to validate our experiment is that the FPR should be inferior to 5% and the TPR should be superior to 90%. We have selected a threshold that produces an $FPR = 2\%$ (arbitrarily). The corresponding threshold value is $Th = 0.585$ and the $TPR = 93.91\%$. This model with this threshold level will thus satisfy our minimum performance criteria. In the following Section, we will evaluate the model with this final threshold in further detail.

This ROC curve can also be used to directly quantify the quality of the classifier. It uses the Area Under the Curve (AUC) as an indicator of high-level performance. The maximum value of the AUC is equal to 1 for a perfect detector. Here, the proposed model has an $AUC = 0.989$ using the complete test dataset. In Table 2, the Areas Under the Curves (AUC) are given for the different categories of data: the contact opening, the carbonised path, 400 Hz, 600 Hz, 800 Hz and the complete database.

Table 2
ROC Area Under the Curve (AUC) for the different categories of test data. CO: Contact Opening. CP: Carbonised Path.

	CO	CP	400 Hz	600 Hz	800 Hz	All
AUC	0.986	0.991	0.990	0.989	0.988	0.989

6.3. The detection performance level at $Th = 0.585$

The confusion matrices in Tables 3, 4, 5 have been obtained by evaluating the model using the complete test database. We have given the overall results and the signal-category results using the carbonised path data, the contact opening data and finally by using both data. The values in the confusion matrices are the percentage of the number of classified current points. It should be noted that, in scientific literature, the performance levels are often presented as a percentage of the number of correctly identified current windows (Wang et al., 2018b; Paul et al., 2021).

We provide a report, in Table 6, on the four usual performance measurements: binary accuracy, precision, recall, false positive rate and F-score.

7. Discussion

7.1. Advantages of the proposed approach

These were our findings: The proposed TNN has a level of 96.3% accuracy and 1.99% FPR using the test database. There is no reference

Table 3
Confusion matrix for carbonised-path scenarios (in %). Total number of points: 437 760.

		Predicted: Arc	Predicted: No arc
Actual	Arc	19.6	2.0
	No arc	0.9	77.5

Table 4
Confusion matrix for the contact-opening scenarios (in %). Total number of points: 583 680.

		Predicted: Arc	Predicted: No arc
Actual	Arc	53.2	2.9
	No arc	1.5	42.3

Table 5
Confusion matrix for both the carbonised-path and contact-opening scenarios (in %). Total number of points: 1 021 440.

		Predicted: Arc	Predicted: No arc
Actual	Arc	38.8	2.5
	No arc	1.2	57.4

Table 6
Detection-performance measurements. FPR: False Positive Rate.

Scenarios/scores (%)	Carbonised path	Contact opening	Both scenarios
Accuracy	97.2	95.5	96.3
Precision	96.5	97.2	97.0
Recall	90.8	94.8	93.9
FPR	1.99	1.99	1.99
F1-score	93.5	96.0	95.4

Table 7
Quantitative synthesis of DL methods applied to AC series-arc-fault detection. Accuracy and FPR are shown only for illustrative purpose. They are not statistically comparable since the models shown in this table do not share the same labelling technique and the same test dataset.

	Deep MLP (Wang et al., 2018b)	CNN (Paul et al., 2021)	RNN (Li et al., 2020a)	Proposed TNN
Accuracy	94.3%	99.36%	94%	98.9%
FPR	0.7%	0.22%	n/a	2%
Number of parameters	3k–250k[a]	79k	53k	2.2k
Temporal resolution	10 ms	10 ms	20 ms	10 µs

[a]This range of number of parameters is estimated using the structure of the neural network.

model on this public database nor on signals of variable frequency power distribution networks (400 Hz, 600 Hz, 800 Hz) to compare our performances with. Perhaps, these performances can be used as reference. However, our private training data comes from the same experiment as the public test data. Thus other models tested on this public data are expected to have significantly lower performances. Contrary to what was expected, the performance is slightly lower using a data type similar to the training data (contact opening) $AUC = 0.986$ and higher using the other data type (carbonised path) $AUC = 0.991$. This performance using a data type, unseen by the TNN, shows that the TNN has also some generalisation capabilities. To our knowledge and, after having reviewed the results presented in scientific literature, the proposed TNN has the lowest number of learning parameters of a DL algorithm that has been applied to the problem of arc-fault detection (2266 parameters versus 53 000 parameters in Li et al. (2020a)) see Table 7. The fact that the number or parameters remains low is promising in terms of hardware implementation as this reduces the amount of resources required (i.e.: processing time and memory). The Table 7 also shows that the proposed model has a temporal resolution one thousand time finer than the shown window-based detection algorithms. We have therefore shown, without the need to design expert features, that a DL algorithm, such as the TNN, can perform arc-fault detection on a *AC* 400 Hz–800 Hz aircraft power distribution current signal. The proposed TNN architecture allows the processing of data segments in parallel (for fast implementation) and the self-attention mechanism has increased ability in terms of capturing time dependencies.

7.2. Limitations of the proposed approach

Despite these advantages, we have identified some limits. One of the first is the detection performance level (in terms of accuracy, recall, precision, FPR and F1-score). This can be explained by a range of different reasons however. The training database is very small: it is the same size as the validation database. Adding more data would generally increase the quality of the performances (Goodfellow et al., 2016).

Most of the arcs that are not successfully identified occur rarely in the training data. The beginning phase of the arc (the first arcing half-period) might be missed because only one beginning phase occurs per training signal, see Fig. 6. The difference between training data distribution and test data distribution is a general problem for DL. In our case, it is of interest to be able to visualise and confirm which events are the hardest to detect because it can help us to select the data that must be added to the training dataset.

Another reason that can explain why the detection results are 96.3%, is that these results are evaluated in relation to an imperfect ground truth label. As is the case with all other labelling techniques, our technique is prone to a level of labelling error which can reduce the quality of the detection results. These errors are known for misleading the training and the evaluation stages.

7.3. Furtherwork

Now that it has been shown how the TNN is relevant to the cause of AC arc-fault detection, we can propose further studies. It will be necessary to build large public databases in order to truly evaluate and compare AI-based arc-fault detectors. In this scenario, it would thus be possible to benchmark all models proposed by the community using the same data. The quality of the databases, notably their fidelity to real world data, should be the object of systematic study. Also studying the role played by the size of ground-truth label (in terms of detection performance and the inference time of the model) would help to choose a detection strategy. This definition of label size could also suggest how to clearly separate the study of arc-fault identification (output probability) from arc-fault detection involving a decision logic. In terms of TNN in particular, it would be interesting to use explainability tools in order to discover the relationship between the performances, the model structure and the data. Finally, studying the hardware implantation of TNNs would help to determine if the parallelism of the attention mechanisms can reduce the detection-inference time (as compared to other models). Several papers on the hardware on implementation have been presented in the literature. Tzanos et al. (2022) proposed a novel architecture and a hardware acceleration using FPGAs. Li et al. (2020b) proposed an acceleration framework, Ftrans, for transformer-based large scale language representations using FPGA. Their implementation is compared to CPU and GPU in terms of energy efficiency. Peng et al. (2021) also described an accelerating transformer on FPGAs and investigates the column balanced block-wise pruning on the model.

Conclusion

We have successfully managed to demonstrate that the TNN can have an elevated accuracy for a given false-positive rate and at a very fine temporal resolution, using a publicly available database (with the most common load). The proposed TNN possesses a very low number of parameters. The parallelism of its attention mechanisms allows a hardware implementation that meets the arc-fault detection memory and time constraints. The high accuracy obtained on a data type unseen in the training demonstrates the generalisation capability of the TNN. This ability to generalise from contact opening to carbonised path scenario shows that the TNN and its attention mechanisms are able to capture a hidden signature common to the two types of signals used for this evaluation. It is therefore the case that TNNs should be included in the selection of promising deep-learning models that can be used to study the process of AC series-arc-fault detection in aircraft. The next logical step would be the production and publication of a series of extensive, and publicly available, arc-fault databases in order to compare the different AI detection models.

CRediT authorship contribution statement

A. Chabert: Writing – original draft, Conceptualization, Methodology. **M.C. Bakkay:** Writing – original draft, Conceptualization, Methodology. **P. Schweitzer:** Data curation, Conceptualization. **S. Weber:** Conceptualization. **J. Andrea:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw experimental validation database called AFDACA is available from Schweitzer et al. (2021) at:
https://zenodo.org/record/5503454#.YpRmM1TP270
For further analysis, the prediction results on the complete test database can be found on github:
https://github.com/alexchabert/electric-arc-detection.

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